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Education

Institute of Science Tokyo (formerly Tokyo Institute of Technology)¹ <i>Doctor of Science in Mathematics</i> Supervisor: Kazuma Shimomoto	Tokyo, Japan <i>Apr 2024–Mar 2026</i>
Tokyo Institute of Technology <i>Master of Science in Mathematics</i> Supervisor: Kazuma Shimomoto (2nd year), Fumiharu Kato (1st year)	Tokyo, Japan <i>Apr 2022–Mar 2024</i>
Tokyo Institute of Technology <i>Bachelor of Science in Mathematics</i> Supervisor: Fumiharu Kato	Tokyo, Japan <i>Apr 2018–Mar 2022</i>

Professional Position

JSPS Research Fellow (PD) <i>Institute of Science Tokyo</i> Host Researcher: Kazuma Shimomoto	Apr 2026–Mar 2027
JSPS Research Fellow (DC1) <i>Tokyo Institute of Technology</i> Host Researcher: Kazuma Shimomoto	Apr 2024–Mar 2026

Research Interests

Commutative algebra in mixed characteristic (via arithmetic methods such as perfectoid rings, prismatic cohomology, and almost mathematics).

Papers and Preprints

14. **R. Ishizuka** and L. Navarro Chafloque, “On perfectoidization of finite algebras over a perfectoid ring”, arXiv:2606.12229, 2026.
13. **R. Ishizuka** and L. Navarro Chafloque, “Decomposition theorem for perfectoid rings along general ideals”, arXiv:2606.06241, 2026.
12. **R. Ishizuka** and S. Miyashita, “Trace ideals of exterior powers of the module of differentials”, arXiv:2605.00418, 2026.
11. **R. Ishizuka** and S. Yoshikawa, “A local-global correspondence for perfectoid purity”, arXiv:2604.25265, 2026.
10. **R. Ishizuka** and S. Yoshikawa, “Algebraization of absolute perfectoidization via section rings”, arXiv:2604.02682, 2026.
9. **R. Ishizuka** and S. Yoshikawa, “Derived graded modules”, arXiv:2601.19164, 2026.
8. **R. Ishizuka** and S. Yoshikawa, “Graded perfectoid rings”, arXiv:2511.02322, 2025.
7. **R. Ishizuka** and K. Shimomoto, “Quasi-canonical lifting of projective varieties in positive characteristic”, arXiv:2506.01345, 2025.

¹In Oct 2024, Tokyo Institute of Technology was merged with Tokyo Medical and Dental University and reorganized into *Institute of Science Tokyo*.

6. **R. Ishizuka**, “A higher algebraic approach to liftings of modules over derived quotients”, arXiv:2503.17964, 2025.
5. **R. Ishizuka**, “Perfectoid towers generated from prisms”, Nagoya Math. J. **261**, e17 (2026). arXiv:2409.15785, 2024.
4. **R. Ishizuka** and K. Nakazato, “Prismatic Kunz’s theorem”, J. Algebra **693**, 732-769 (2026). arXiv:2402.06207, 2024.
3. D. Dine and **R. Ishizuka**, “Tilting and untilting for ideals in perfectoid rings”, Math. Z. **307**, 66 (2024). arXiv:2308.09600, 2023.
2. **R. Ishizuka**, “A calculation of the perfectoidization of semiperfectoid rings”, Nagoya Math. J. **255**, 742-759 (2024). arXiv:2305.07916, 2023.
1. K. Shimomoto and **R. Ishizuka**, “A mixed characteristic analogue of the perfection of rings and its almost Cohen-Macaulay property”, arXiv:2303.13872, 2023.

Talks

40. Oct 2026. “TBA”, Kinokuni Algebraic Geometry Symposium 2026, Kinokuni International Arts Center, Japan.
39. Sep 2026. “Local and global aspects of perfectoid pure singularities”, The 12th Japan-Vietnam Joint Seminar on Commutative Algebra, Khanh Hoa University, Vietnam.
38. July 2026. “TBA”, Honjo Waseda Algebraic Geometry Seminar 2026, Honjo seminar house, Japan.
37. June 2026. “Graded modules and comodules: The dual notion of modules”, The 36th Seminar on Commutative Algebra in Japan, Chiba University, Japan.
36. Dec 2025. “Global singularities in mixed characteristic and graded perfectoid rings”, Mini-workshop on Arithmetic Geometry in Sendai, Tohoku University, Japan.
35. Nov 2025. “Perfectoid towers arising from Frobenius lifts”, The 46th Japan Symposium on Commutative Algebra, Hotel Fukuracia Osaka Bay, Japan.
34. Nov 2025. “Perfectoid towers arising from prisms”, Workshop on Number theory at Tsuda University 2025, Tsuda University, Japan.
33. Oct 2025. “Absolute perfectoidization of schemes and local to global principle in mixed characteristic”, Singularity Theory Seminar, Nihon University, Japan.
32. Sep 2025. “The Frobenius-Witt cotangent bundle” (Survey talk), Yatsugatake Workshop, 2025, Wellness Garden in Goddess Forest : Kobuchizawa, Japan.
31. Sep 2025. “A version of Kunz’s theorem in mixed characteristic”, Noda Algebraic Geometry Symposium, Tokyo University of Science, Japan.
30. July 2025. “On the derived deformation functor of Frobenius liftings” (Poster Session), 2025 Summer Research Institute in Algebraic Geometry, Colorado State University, USA.
29. July 2025. “On computation of Tor modules over a derived quotient”, The 36th Seminar on Commutative Algebra in Japan., Kyushu University, Japan.
28. June 2025. “A unified construction of perfectoid towers by prisms”, p -adic and Characteristic p Methods in Algebraic Geometry., EPFL, Switzerland.
27. May 2025. “Approximation of perfectoid rings by Noetherian rings and prisms”, MPIM Algebra Seminar., MPIM, Germany.
26. Mar 2025. “On the vanishing of Ext and liftings of modules on derived quotients using higher algebras”, MSJ Spring Meeting 2025., Waseda University, Japan.
25. Mar 2025. “Frobenius liftability and derived algebraic geometry”, The 29th Conference on Algebra for Young Researchers in Japan., Osaka University, Japan.

24. Mar 2025. *"Liftability results from formal moduli problems"* (including survey), Small workshop on arithmetic geometry 2025 in Hakodate., Hakodate Community Design Center, Japan.
23. Nov 2024. *"Prismatic Kunz's theorem"*, The 45th Japan Symposium on Commutative Algebra., RIMS (Kyoto University), Japan.
22. Nov 2024. *"Applications of higher algebra to the lifting problem of modules"*, Tokyo Commutative Algebra Seminar., Online (Zoom), Japan.
21. Oct 2024. *"F-liftability obstruction in singular varieties through derived algebraic geometry"*, Singularity Theory Seminar., Nihon University, Japan.
20. Sep 2024. *"Frobenius maps on mixed characteristic rings via prismatic cohomology"* (Poster Session), L-functions and Motives in Niseko 2024., Setsu Niseko and Niseko Residents Center, Japan.
19. Sep 2024. *"A generalization of Kunz's theorem to mixed characteristic via p-adic cohomology theory"*, MSJ Autumn Meeting 2024., Osaka University, Japan.
18. July 2024. *"Perfectoid spaces, tilts and untilts"* (Survey talk), Atelier de Géométrie Arithmétique 2024., RIMS (Kyoto University), Japan.
17. July 2024. *"Prismatic approach to a mixed characteristic Kunz's theorem"*, The 23rd Hiroshima-Sendai Workshop on Number Theory at Sendai., Tohoku University, Japan.
16. June 2024. *"Regularity criterion of mixed characteristic rings via prismatic cohomology"*, Keio Algebra Seminar., Keio University, Japan.
15. June 2024. *"Prisms and regular local rings"*, The 35th Seminar on Commutative Algebra in Japan., Tokushima University, Japan.
14. Apr 2024. *"Prisms and its application to regular rings"*, Saturday Seminar., Meiji University, Japan.
13. Mar 2024. *"Perfectoid ideals and its correspondence"*, The 20th Mathematics Conference for Young Researchers., Hokkaido University, Japan.
12. Feb 2024. *"Mixed characteristic Kunz's theorem with prismatic theory"*, The 28th Conference on Algebra for Young Researchers in Japan., Waseda University, Japan.
11. Dec 2023. *"Commutative ring theoretic approach for the perfectoidization of semiperfectoid rings"*, Number Theory Seminar., Kyoto University, Japan.
10. Nov 2023. *"Ideal correspondence between a perfectoid ring and its tilt"*, The 44th Japan Symposium on Commutative Algebra., Lecture Hayama, Japan.
9. Aug 2023. *"Absolute integral closure"* (Survey talk), The 18th Summer School on Commutative algebra., Tokyo Institute of Technology, Japan.
8. Aug 2023. *"On the relation between perfectoidization and p-root closure"*, The 9th China-Japan-Korea International Conference on Ring and Module Theory., Incheon National University, Republic of Korea.
7. July 2023. *"On the commutative ring-theoretic structure of the perfectoidization of semiperfectoid rings"*, The 22nd Hiroshima-Sendai Workshop on Number Theory at Hiroshima., Hiroshima University, Japan.
6. July 2023. *"On the application of perfectoidization to commutative algebra and its structure"*, The 34th Seminar on Commutative Algebra in Japan., Kitami Institute of Technology, Japan.
5. May 2023. *"On Perfectoid(ization) and its commutative ring-theoretic properties"*, Ookayama Youth Seminar in Algebra., Tokyo Institute of Technology, Japan.
4. Mar 2023. *"A mixed characteristic analogue of the perfection of rings"*, The 11th Japan-Vietnam Joint Seminar on Commutative Algebra - by and for young mathematicians -, Vietnam Academy of Science and Technology, Vietnam.

3. Mar 2023. *"An explicit construction of perfectoid almost Cohen-Macaulay algebras in mixed characteristic"*, MSJ Spring Meeting 2023., Chuo University, Japan.
2. Mar 2023. *"An explicit construction of perfectoid almost Cohen-Macaulay algebras"*, The 19th Mathematics Conference for Young Researchers., Hokkaido University, Japan.
1. Nov 2022. *"An explicit construction of perfectoid almost Cohen-Macaulay algebras in mixed characteristic"*, The 43rd Japan Symposium on Commutative Algebra., Osaka University, Japan.

Membership

- Apr 2023– . Mathematical Society in Japan (MSJ)

Languages

Japanese: Native

Mother tongue

English: Intermediate

Can read, write, and, listen but may struggle with conversation

French: Beginner

Can only read mathematical texts